SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: Data Visualization with Power BI and Tableau

PRACTICAL NO. 9

AIM: Cleaning and Structuring Data for Analysis

a) Identify and handle missing or inconsistent data

b) Convert wide data into tall format

c) Perform union of multiple files

d) Execute cross-database joins

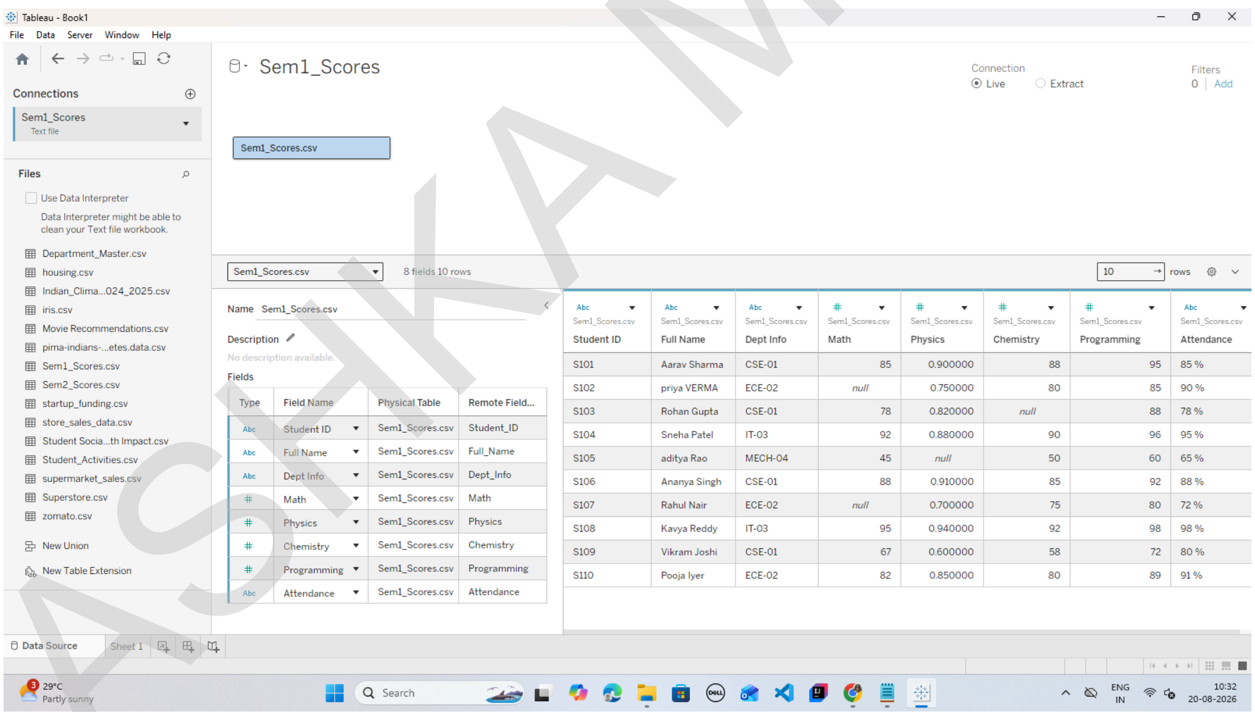
e) Restructure data for better visualization

f) Handle different levels of detail in datasets

g) Apply advanced data cleaning techniques

A: Identify and Handle Missing or Inconsistent Data

Step 1: Load the Dataset



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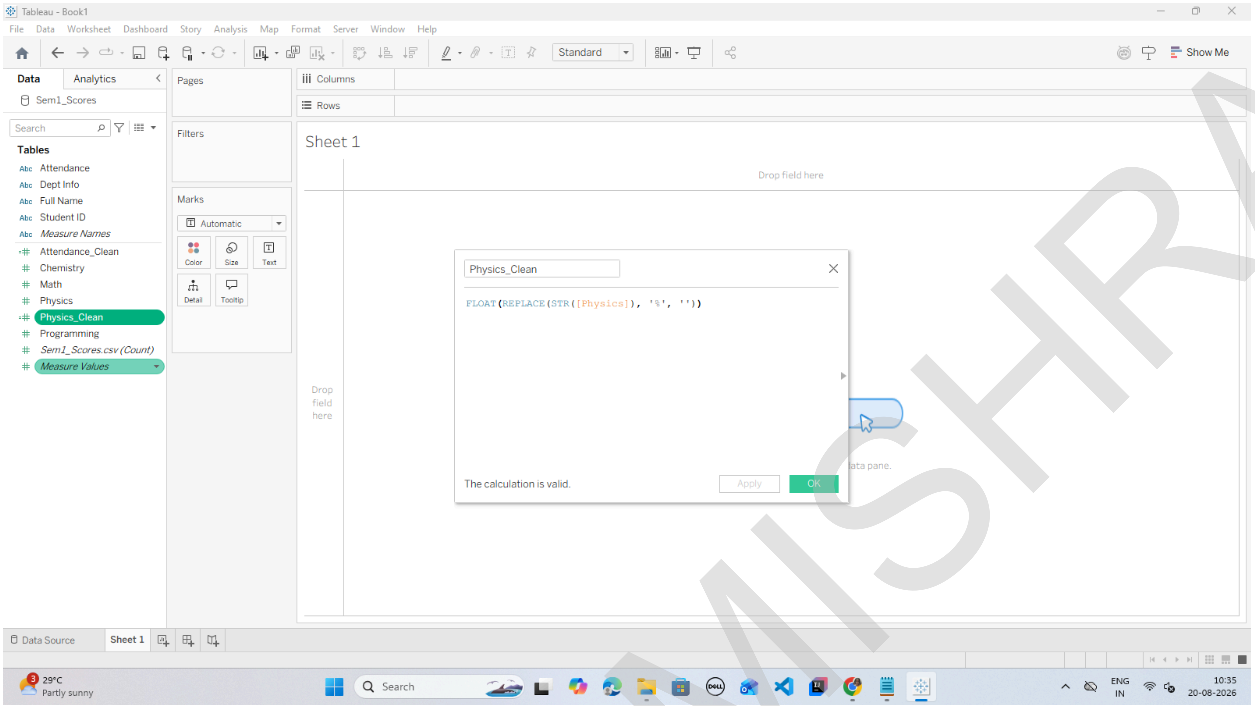
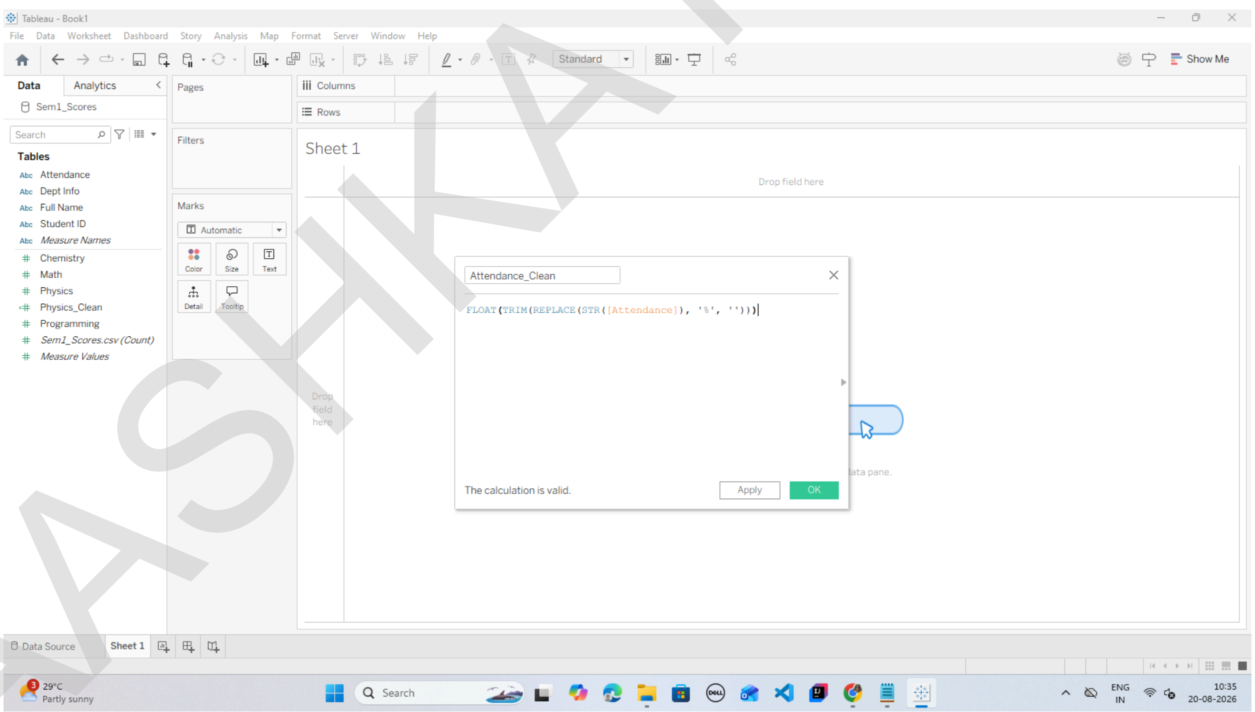
Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric Measures)





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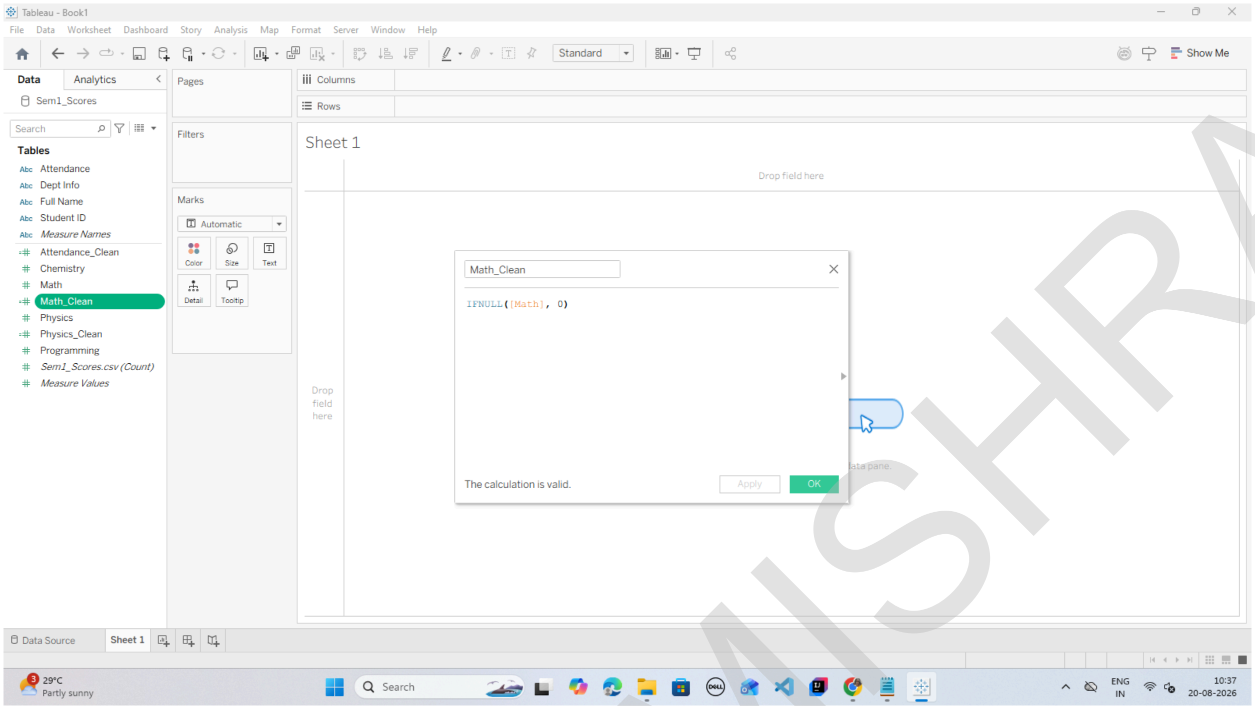
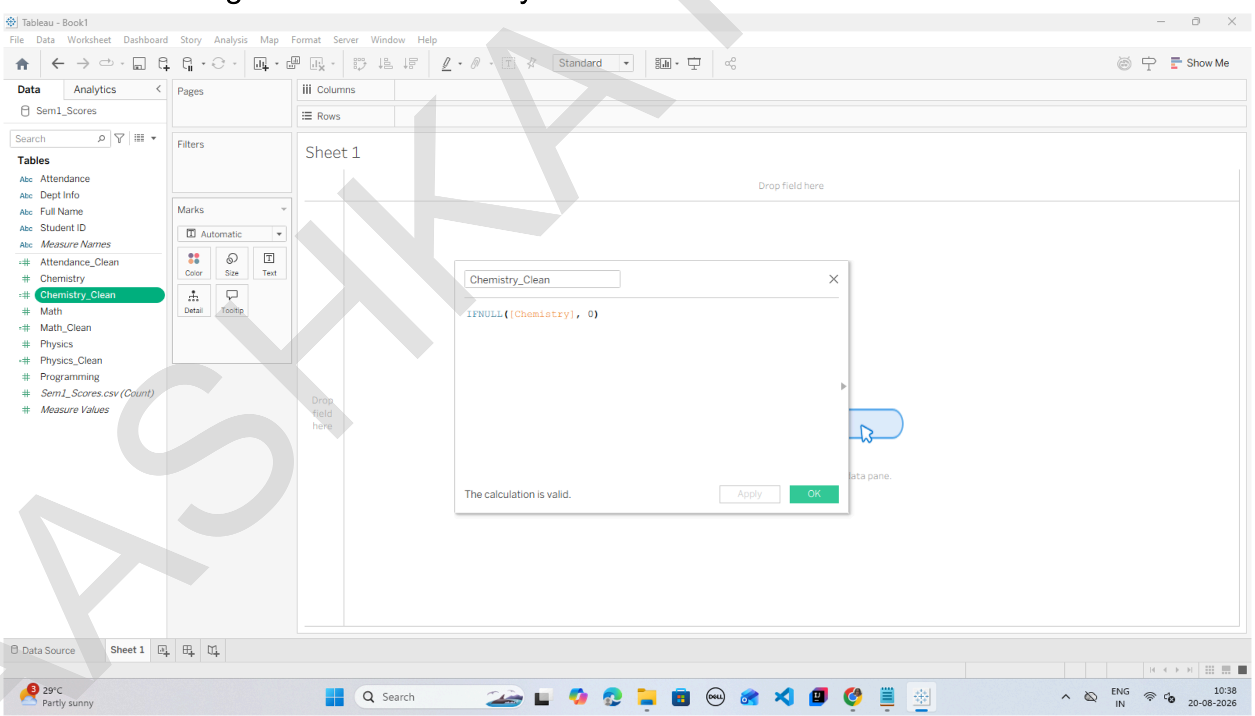
SUBJECT: Data Visualization with Power BI and Tableau

Step 3: Handle Missing (Null) Numeric Values via Imputation



i. Handle missing entries in Chemistry



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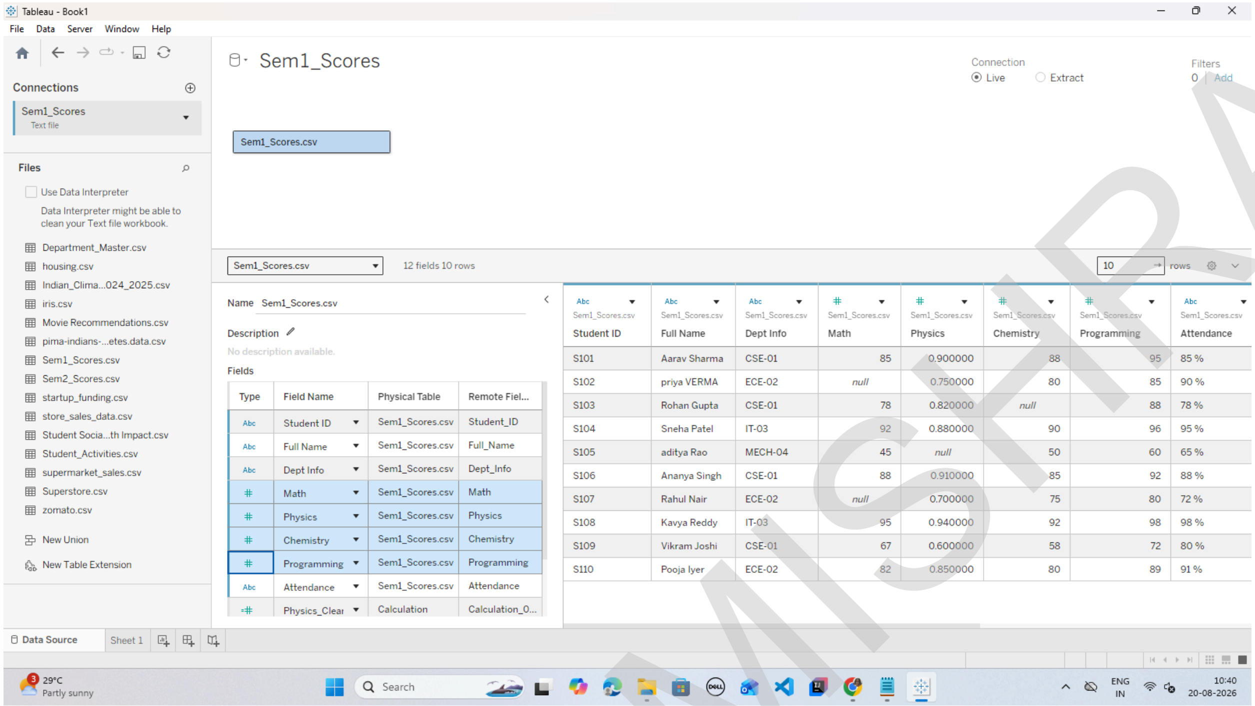
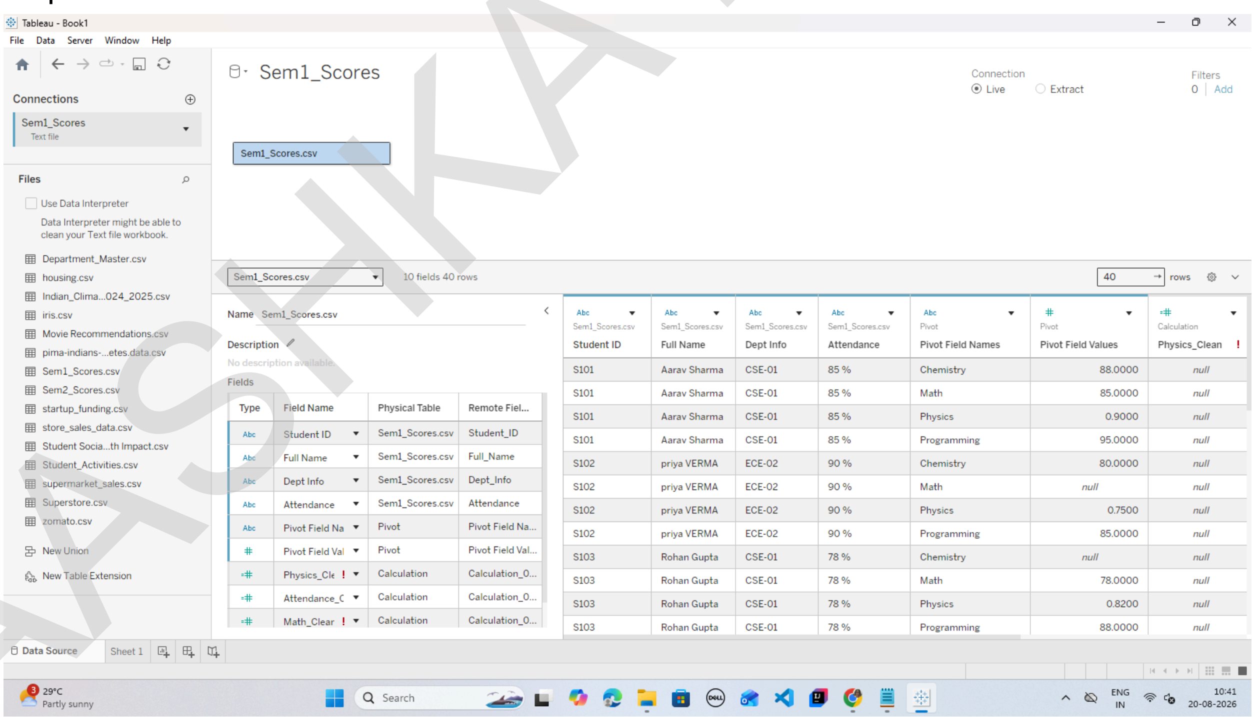
B: Convert Wide Data into Tall Format

Step 1: Open the Data Source Grid



Step 2: Multi-Select and Pivot Columns



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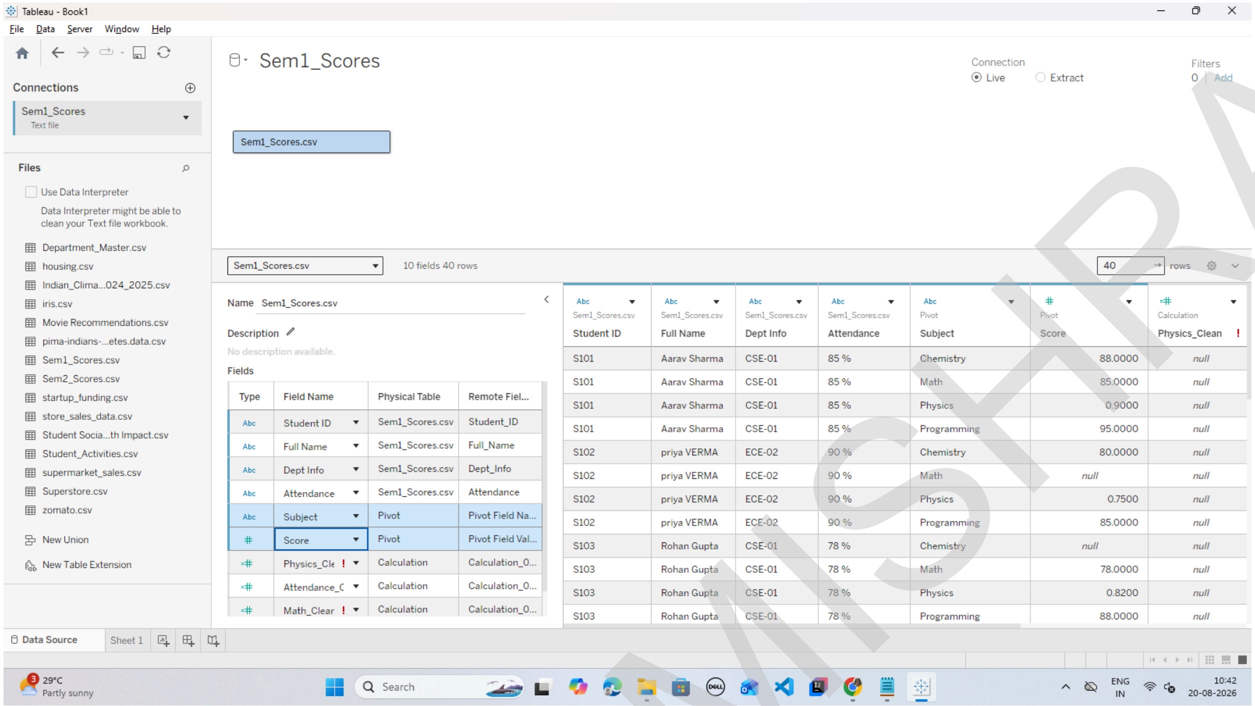
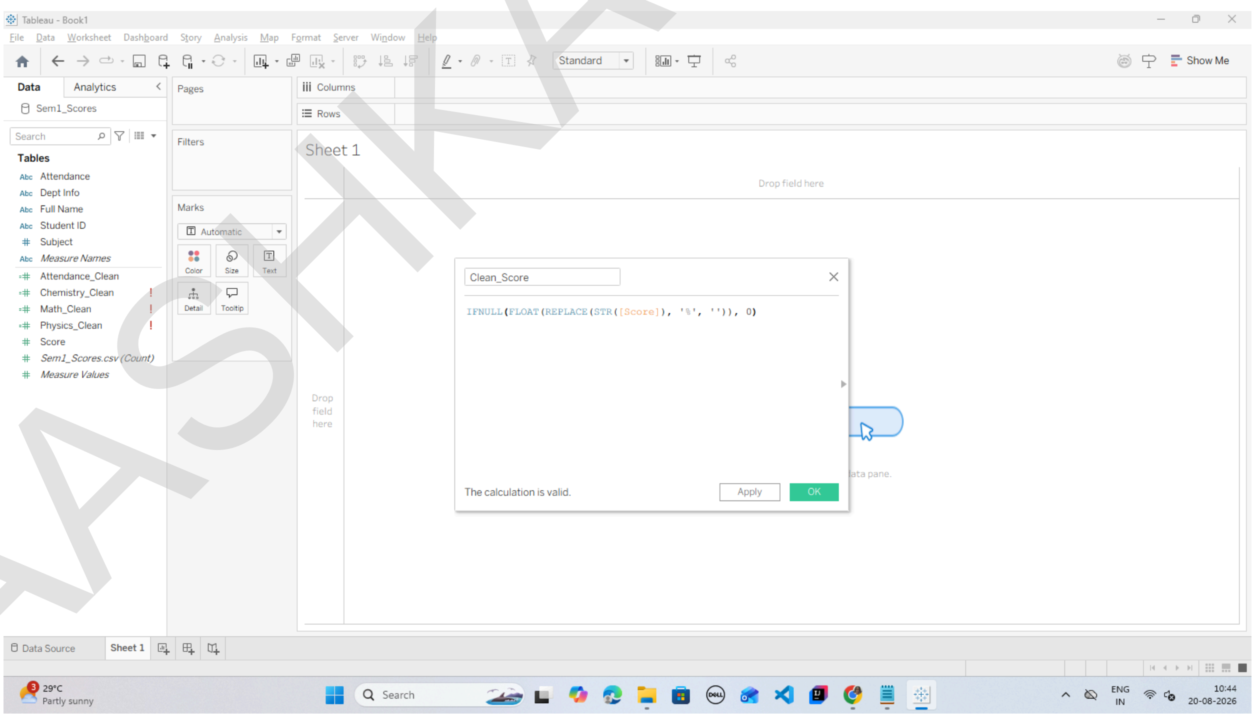
Step 3: Rename the Pivoted Fields Tableau replaces the four columns with two consolidated columns at the end of the table:



Step 4: Clean and Format the Tall Score Field Because raw Physics contained % symbols, the new combined Score column defaults to text (Abc). Convert it to a numeric measure:



NAME: AASHKA MISHRA
ROLL NO: T095

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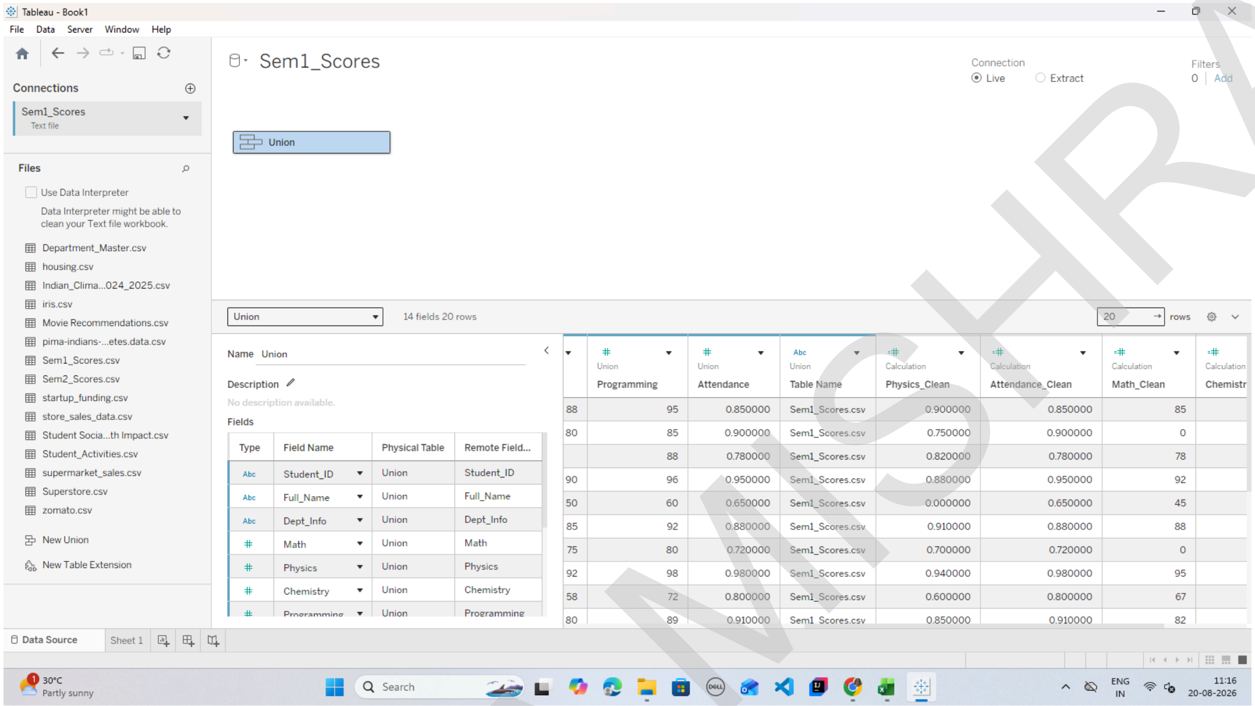
SUBJECT: Data Visualization with Power BI and Tableau

C: Perform Union of Multiple Files

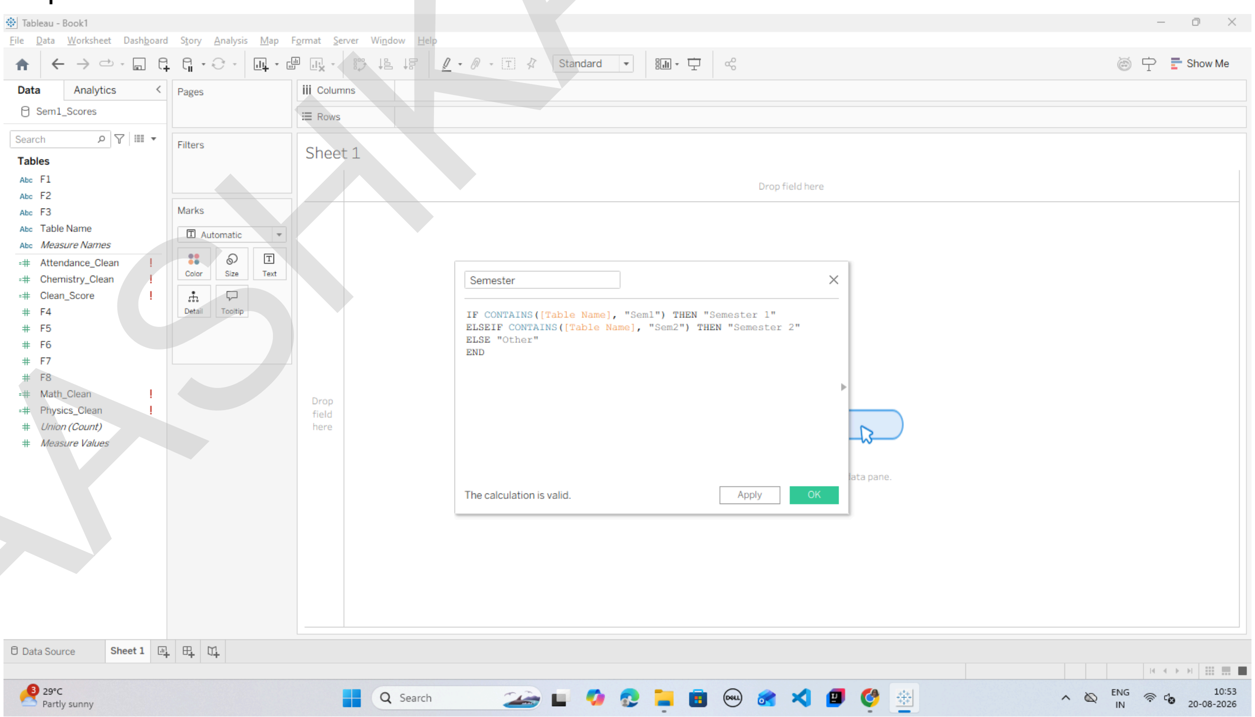
Step 1: Open the Data Source Canvas

Step 2: Create a Multi-File Union

Step 3: Verify the Merged Data Grid



Step 4: Create a Clean Semester Identifier Dimension



NAME: AASHKA MISHRA
ROLL NO: T095

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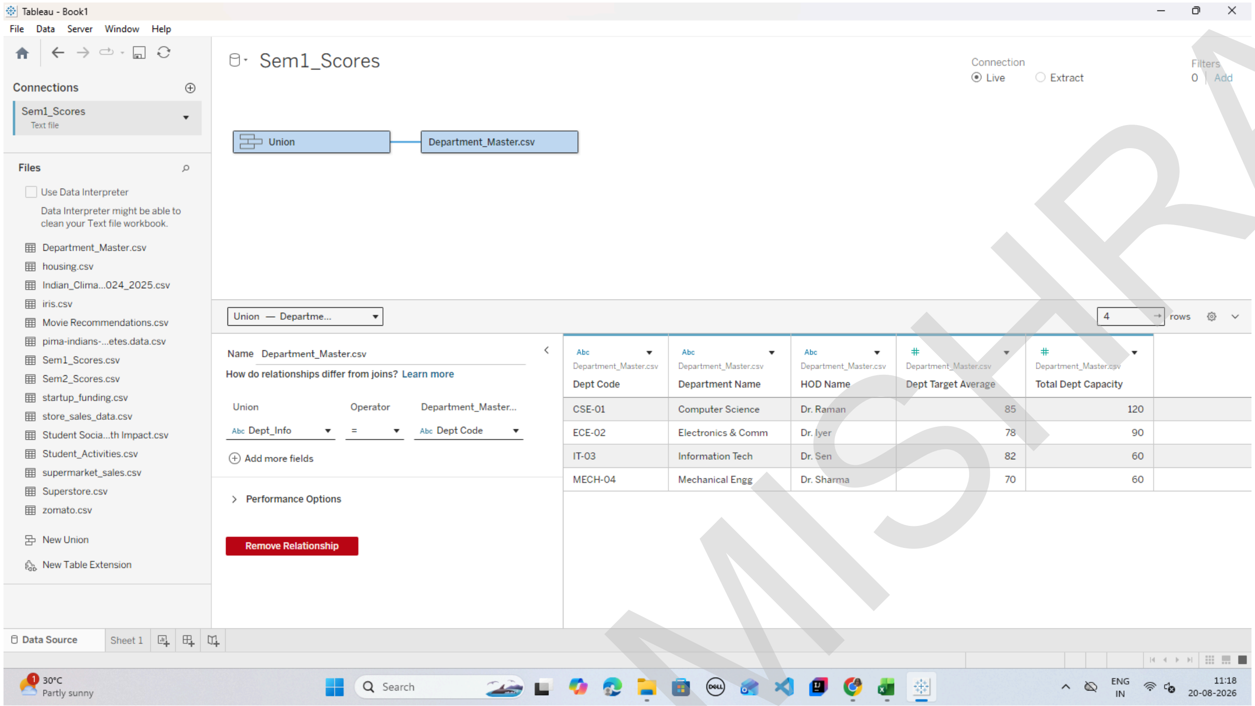
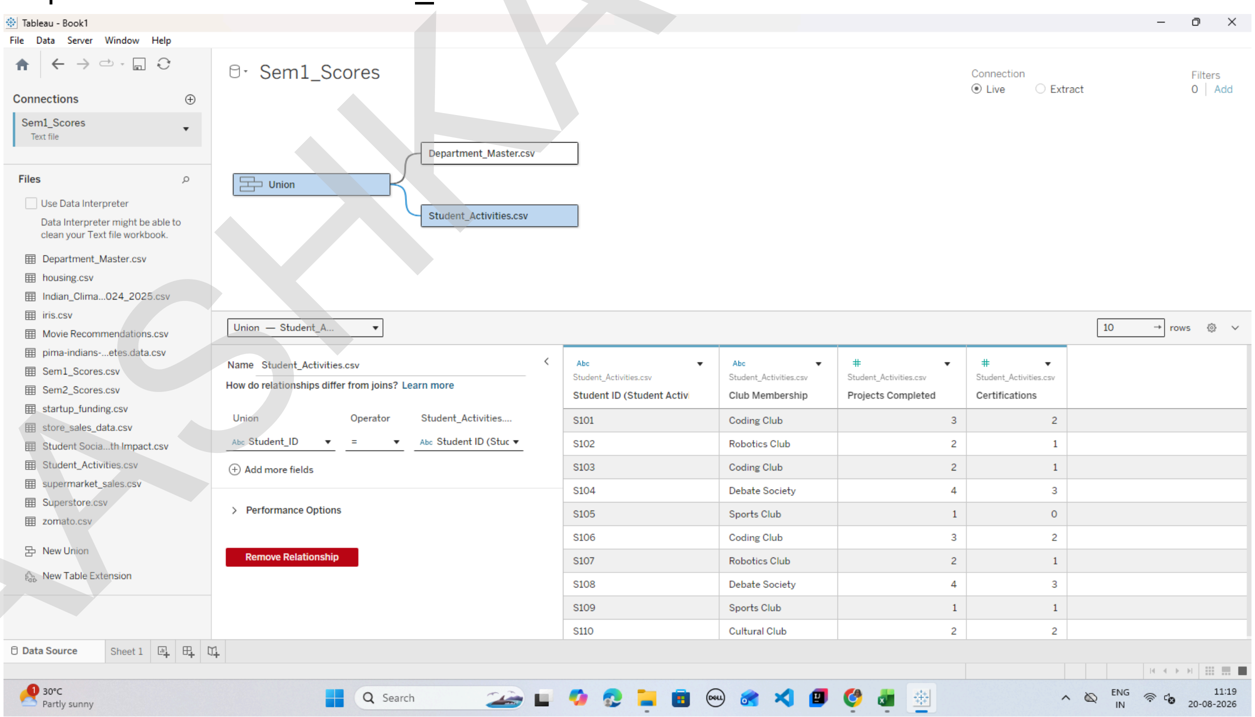
SUBJECT: Data Visualization with Power BI and Tableau

5. D: Execute Cross-Database Joins
Step 1: Stay on the Top Canvas
Step 2: Add and Link Department_Master.csv



Step 3: Add and Link Student_Activities.csv



NAME: AASHKA MISHRA
ROLL NO: T095

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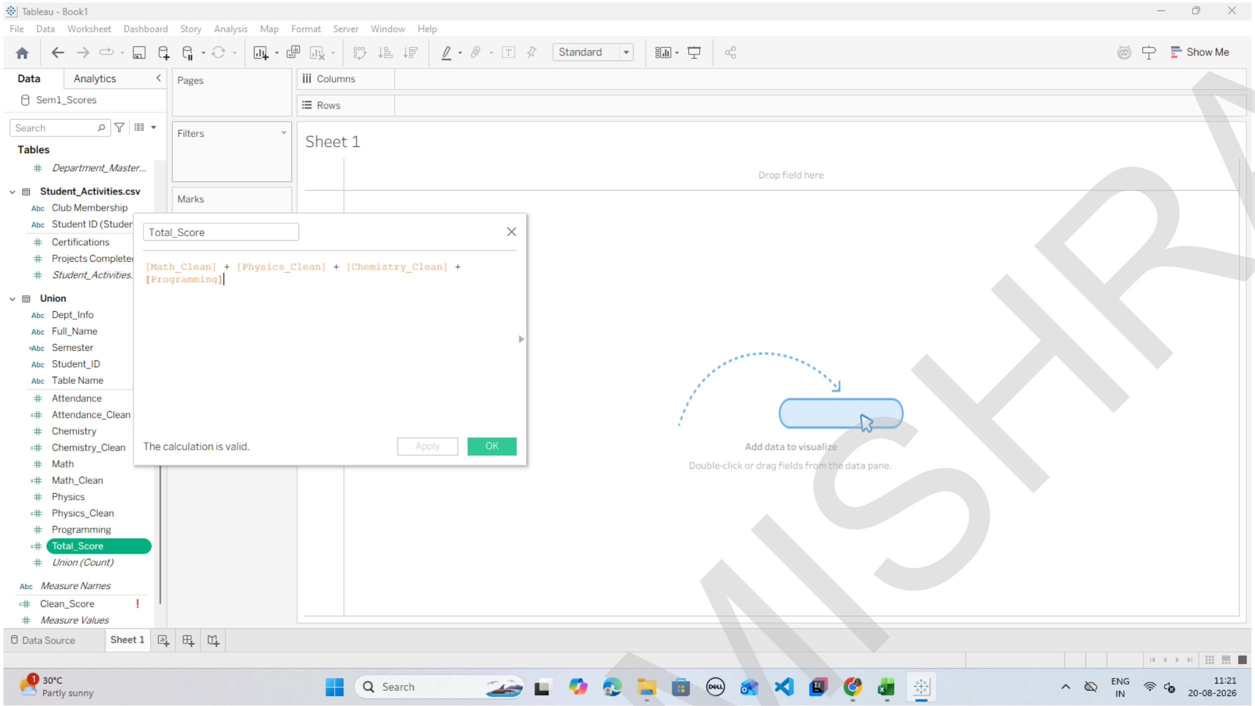
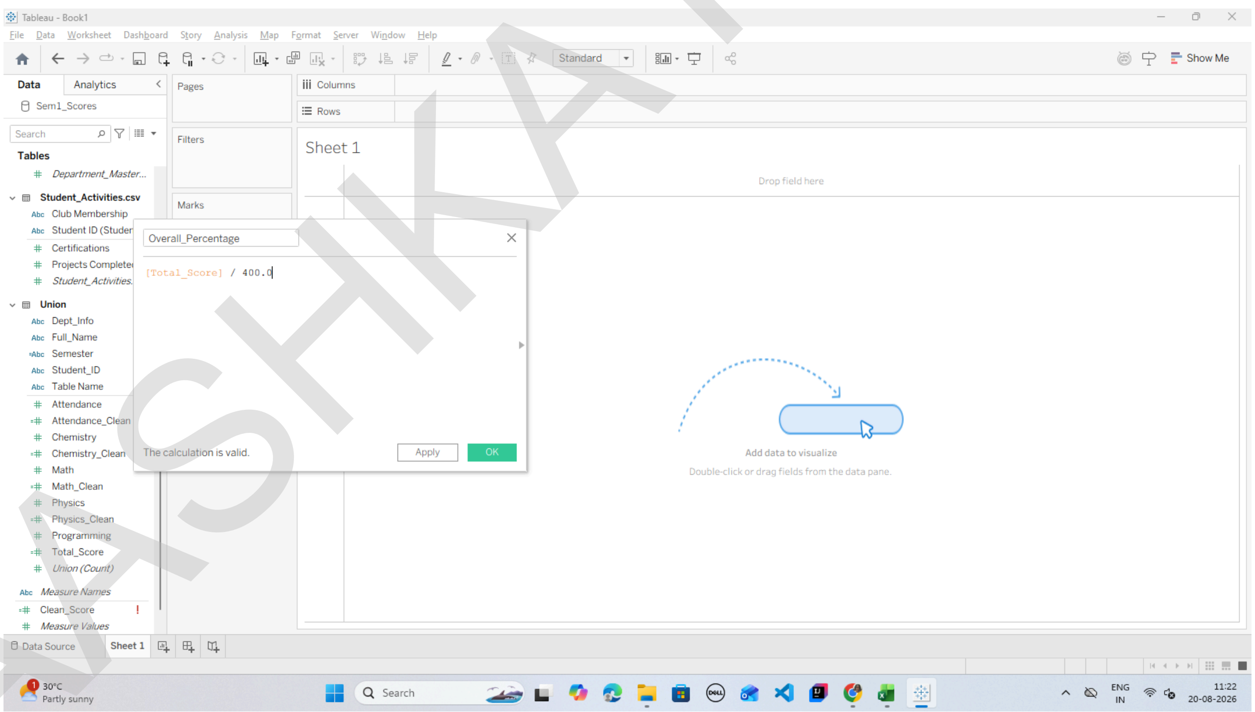
SUBJECT: Data Visualization with Power BI and Tableau

E: Restructure Data for Better Visualization

Step 1: Calculate Total Marks and Scaled Percentage





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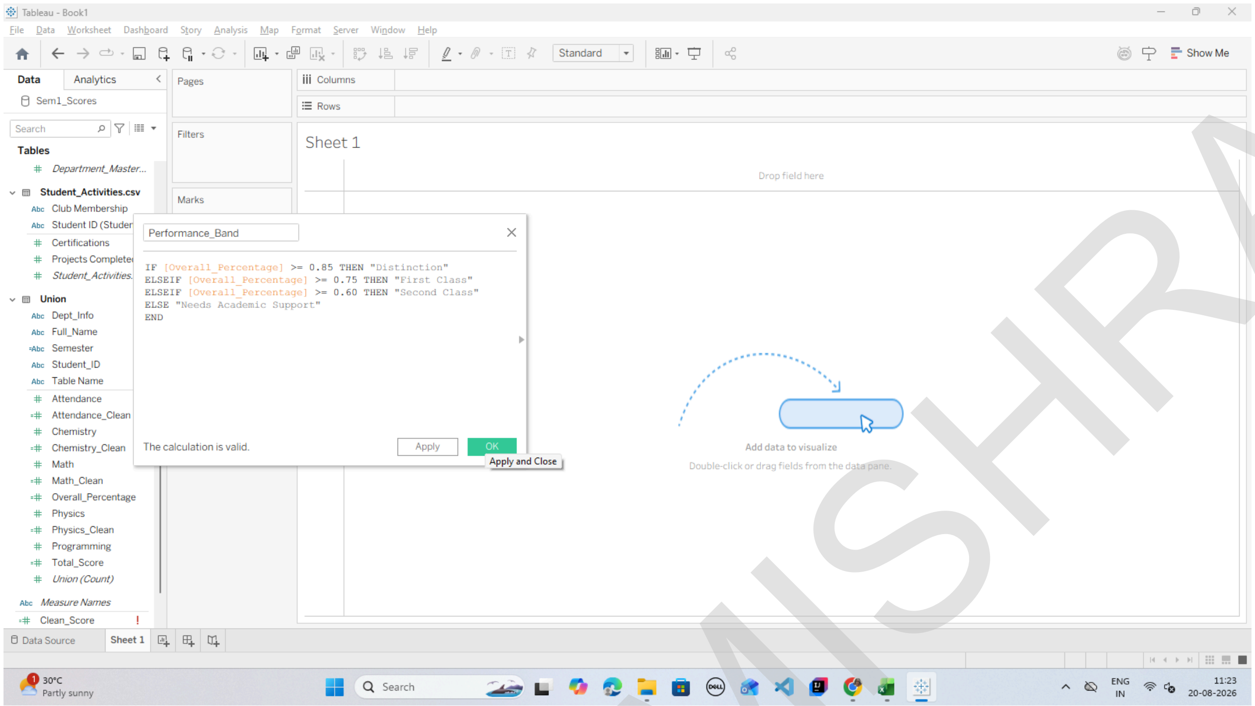
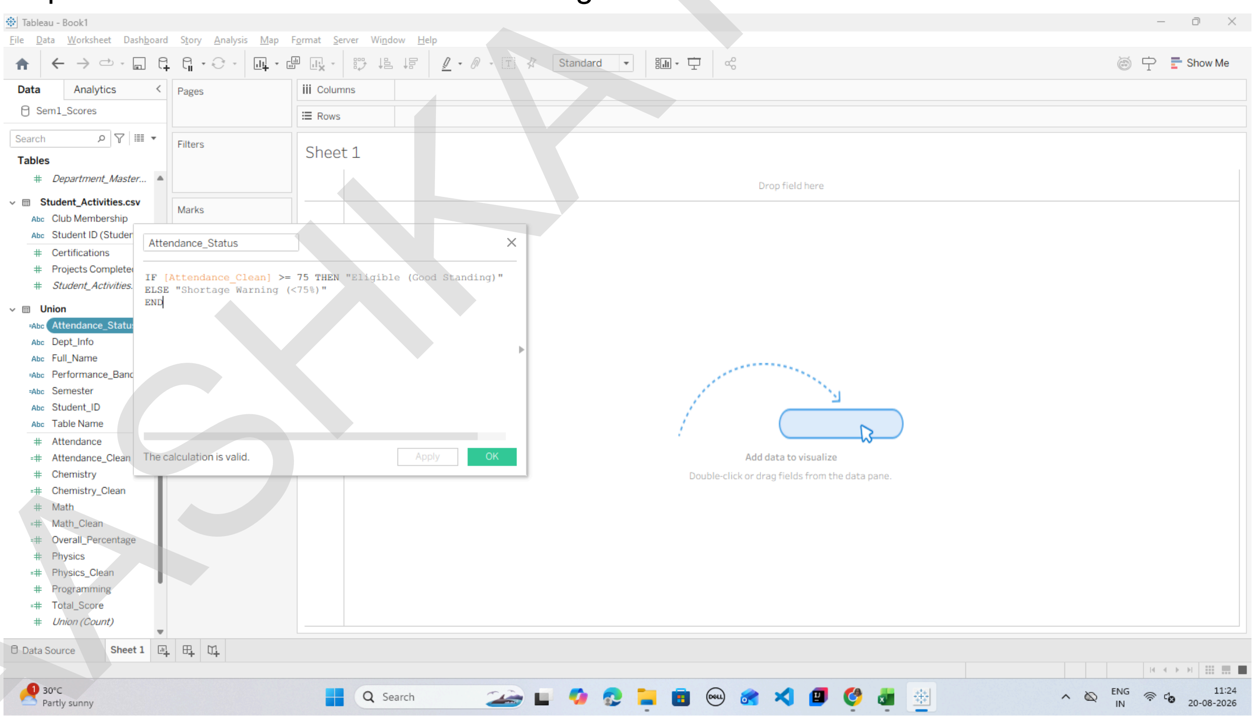
ROLL NO: T095

Step 2: Create Categorical Performance Bands (Grading Tiers)



Step 3: Create an Attendance Health Flag



NAME: AASHKA MISHRA
ROLL NO: T095

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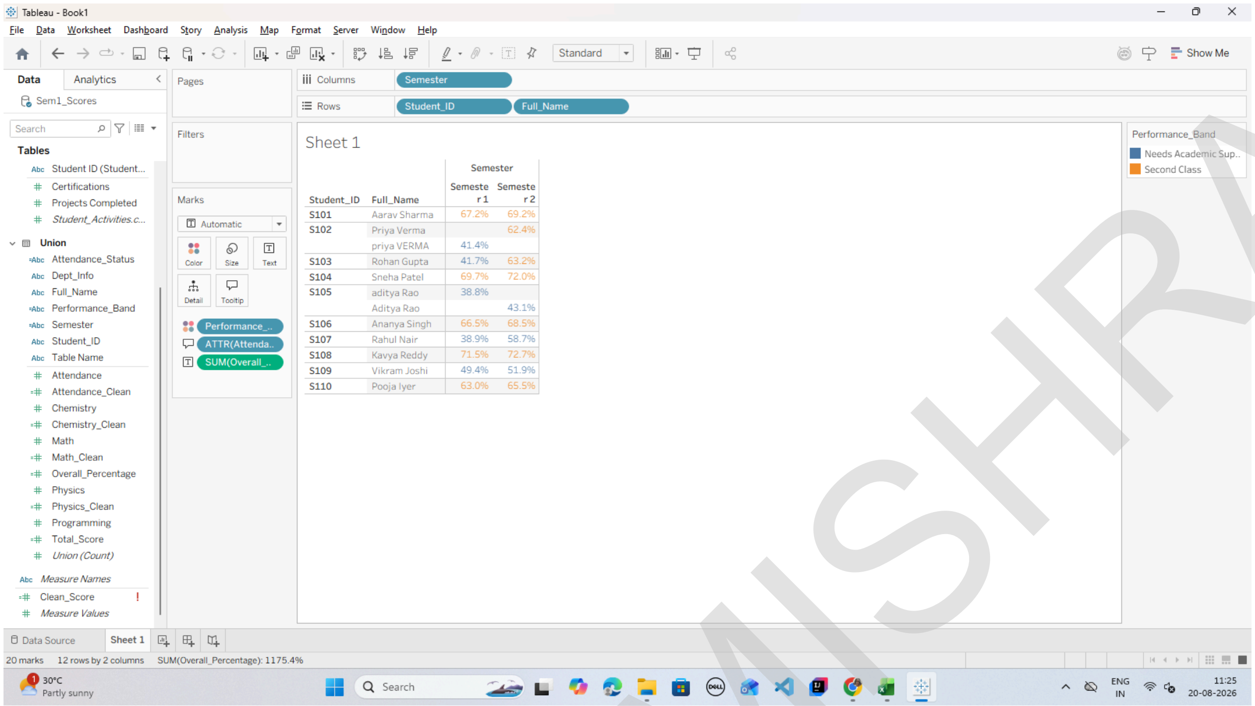
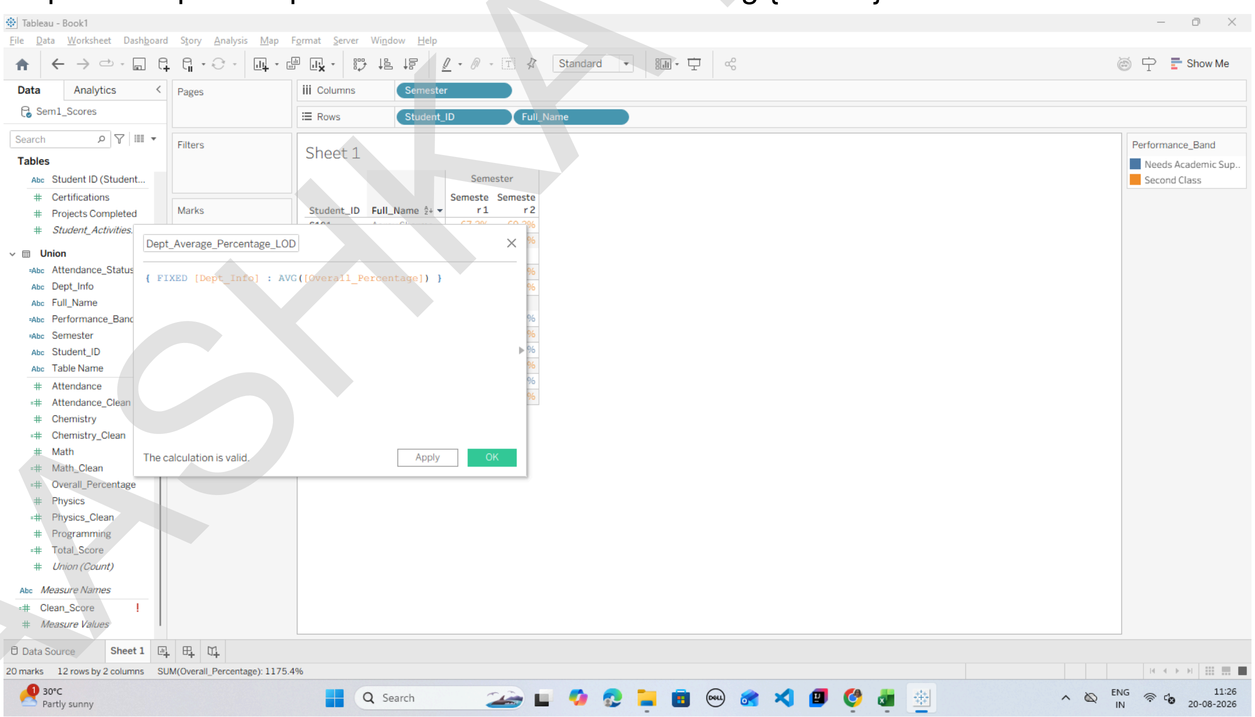
SUBJECT: Data Visualization with Power BI and Tableau

Step 4: Build the Student Performance Scorecard View



F: Handle Different Levels of Detail in Datasets

Step 1: Compute Department-Level Benchmark using {FIXED} LOD



NAME: AASHKA MISHRA
ROLL NO: T095

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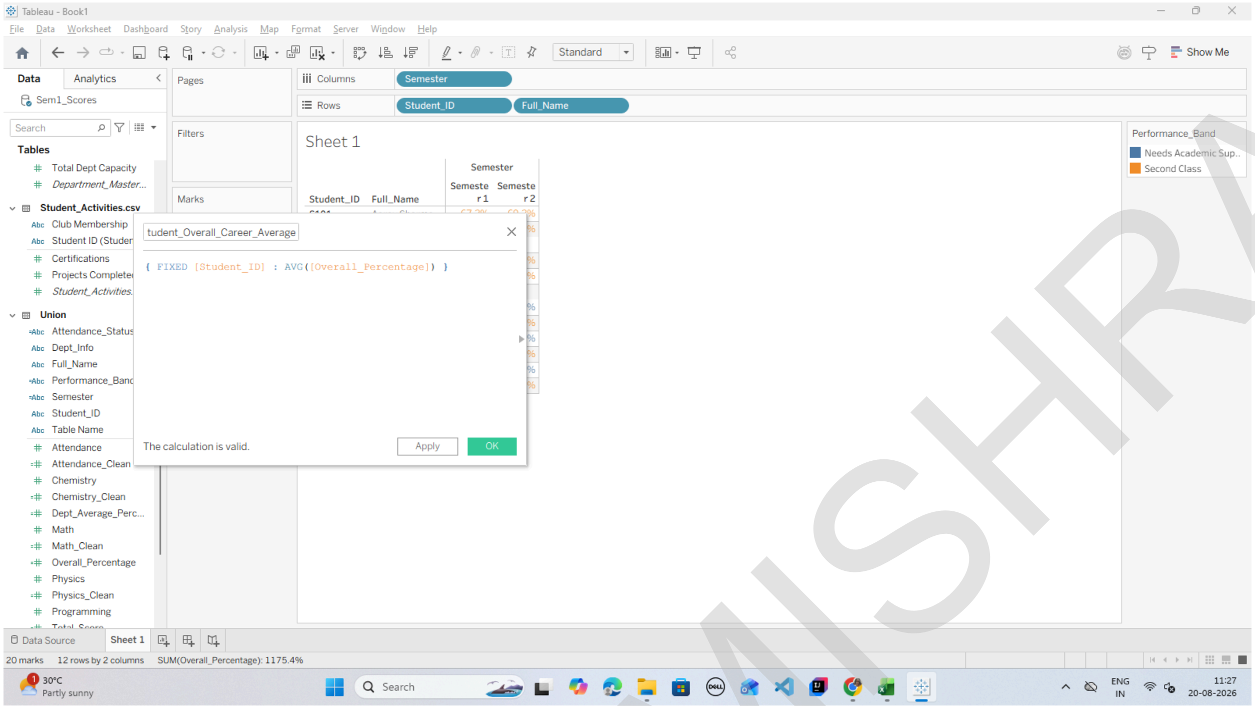
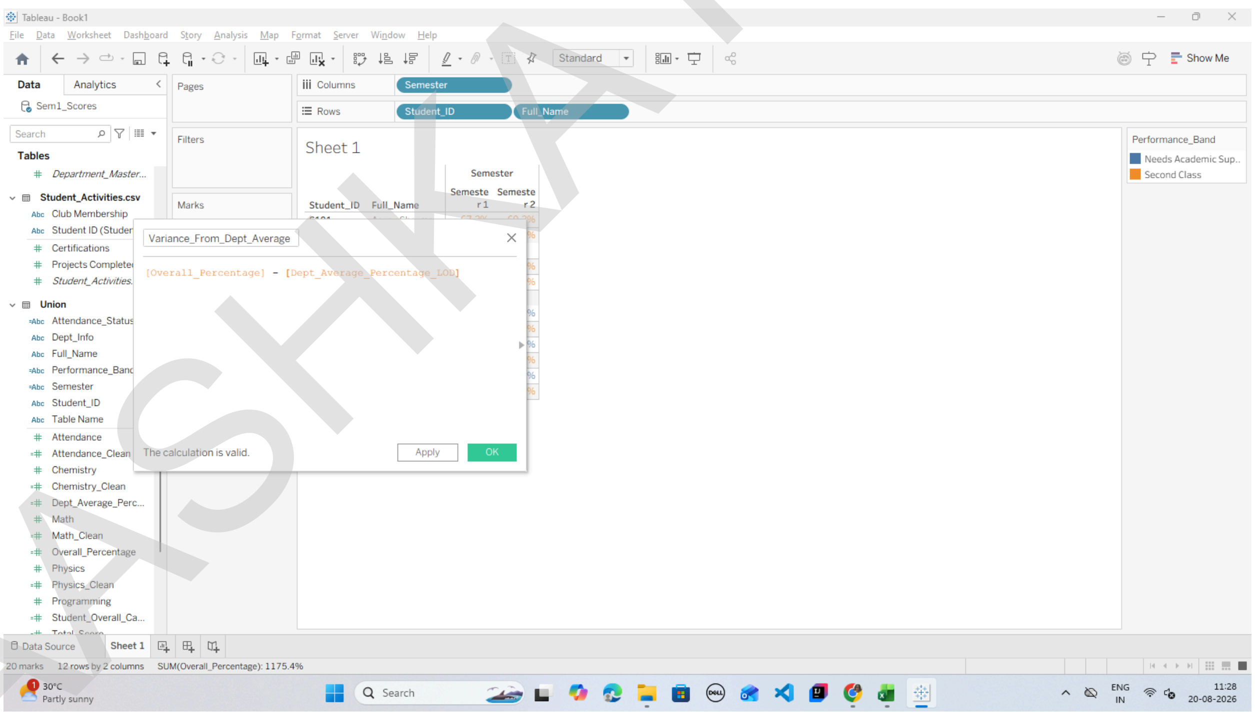
SUBJECT: Data Visualization with Power BI and Tableau

Step 2: Compute Student Multi-Semester Average using {FIXED} LOD



Step 3: Calculate Performance Variance (Student vs. Department Benchmark)



NAME: AASHKA MISHRA
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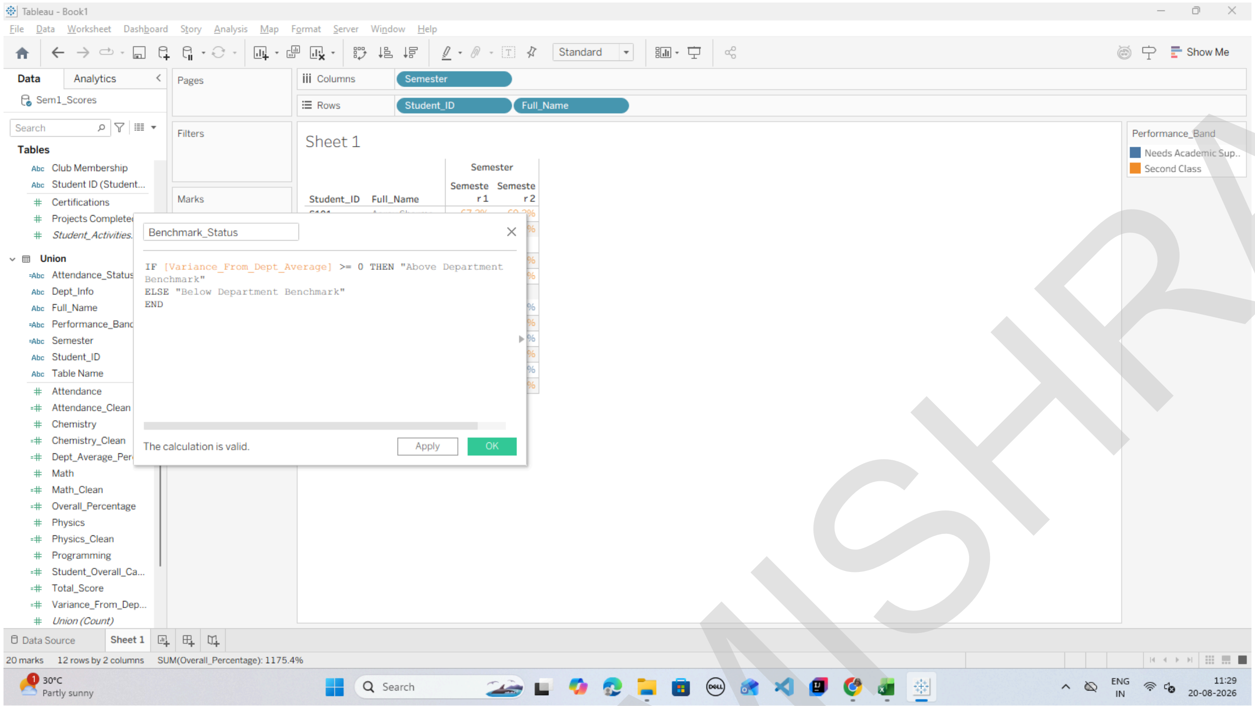
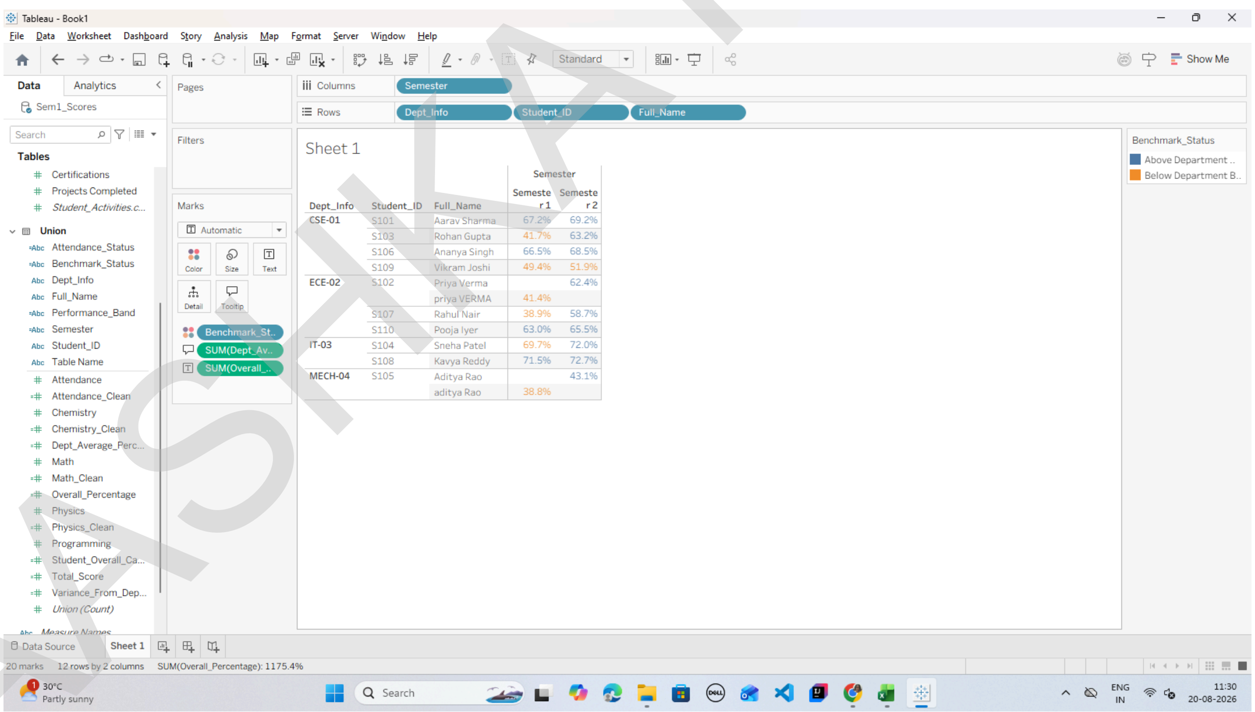
SUBJECT: Data Visualization with Power BI and Tableau

Step 4: Create a Benchmark Indicator Dimension



Step 5: Build the LOD Benchmark Visualization



NAME: AASHKA MISHRA
ROLL NO: T095

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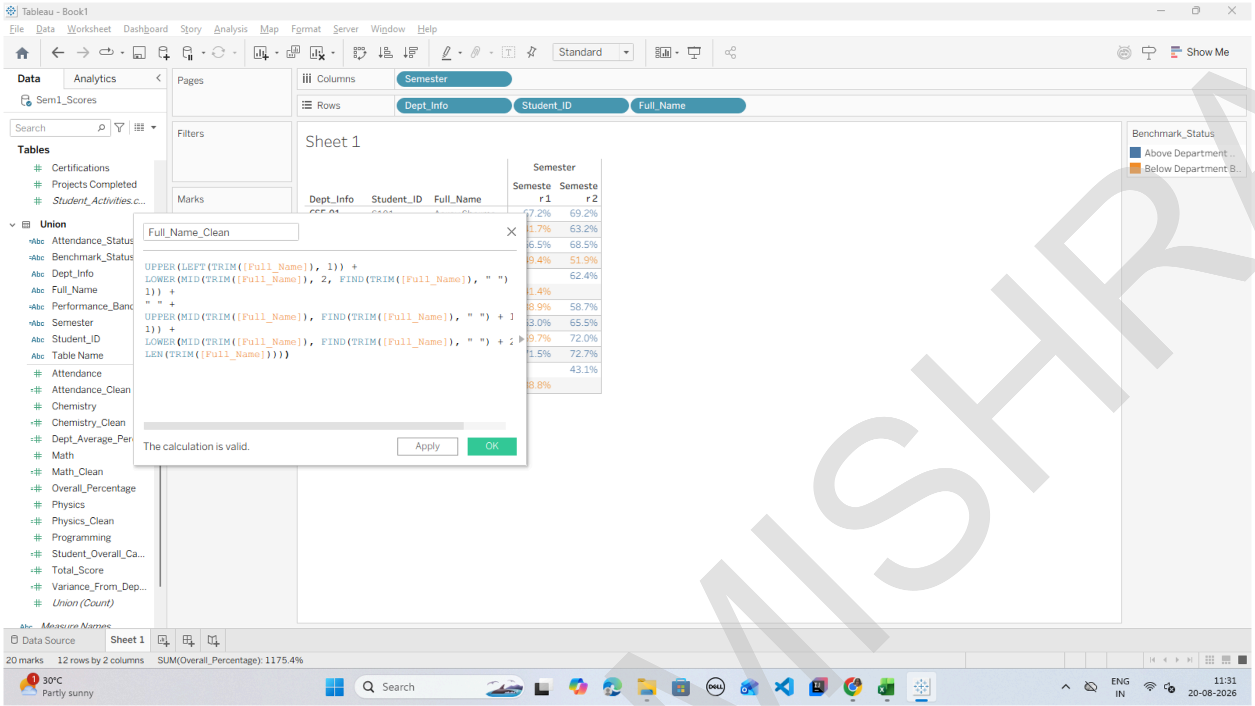
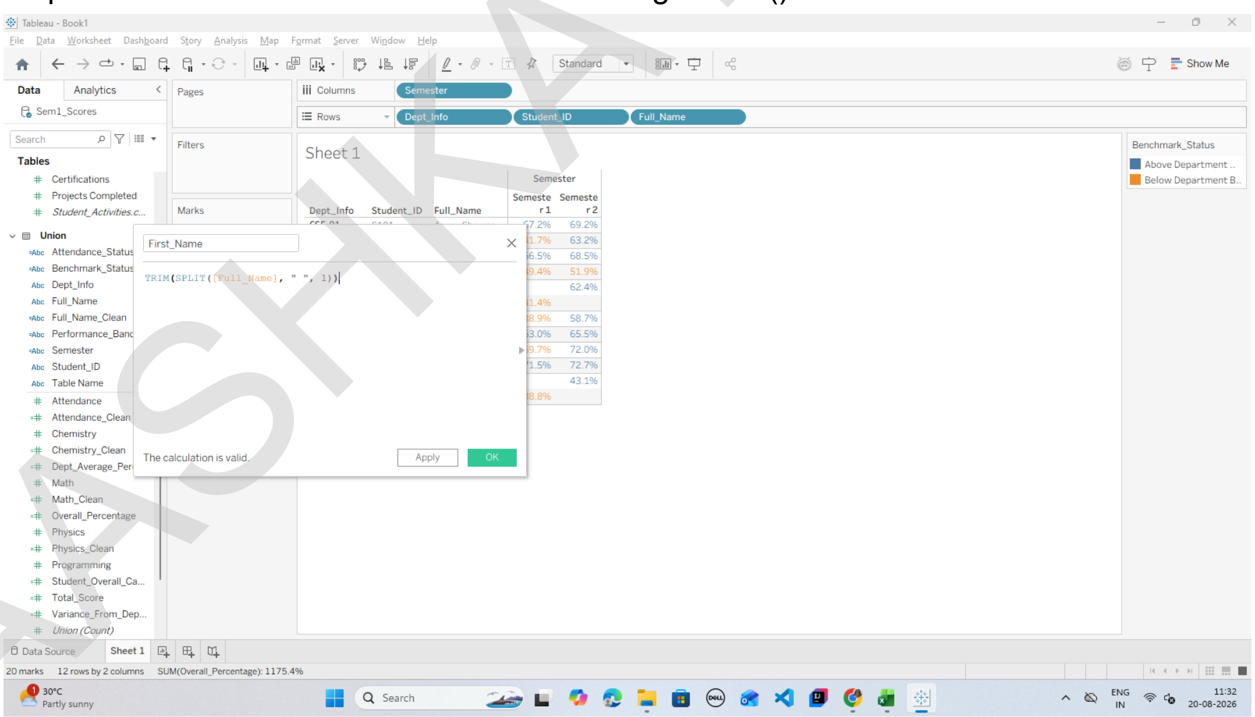
SUBJECT: Data Visualization with Power BI and Tableau

G: Apply Advanced Data Cleaning Techniques

Step 1: Clean Leading/Trailing Whitespace and Inconsistent Name Casing



Step 2: Parse First Name and Last Name using SPLIT()



NAME: AASHKA MISHRA

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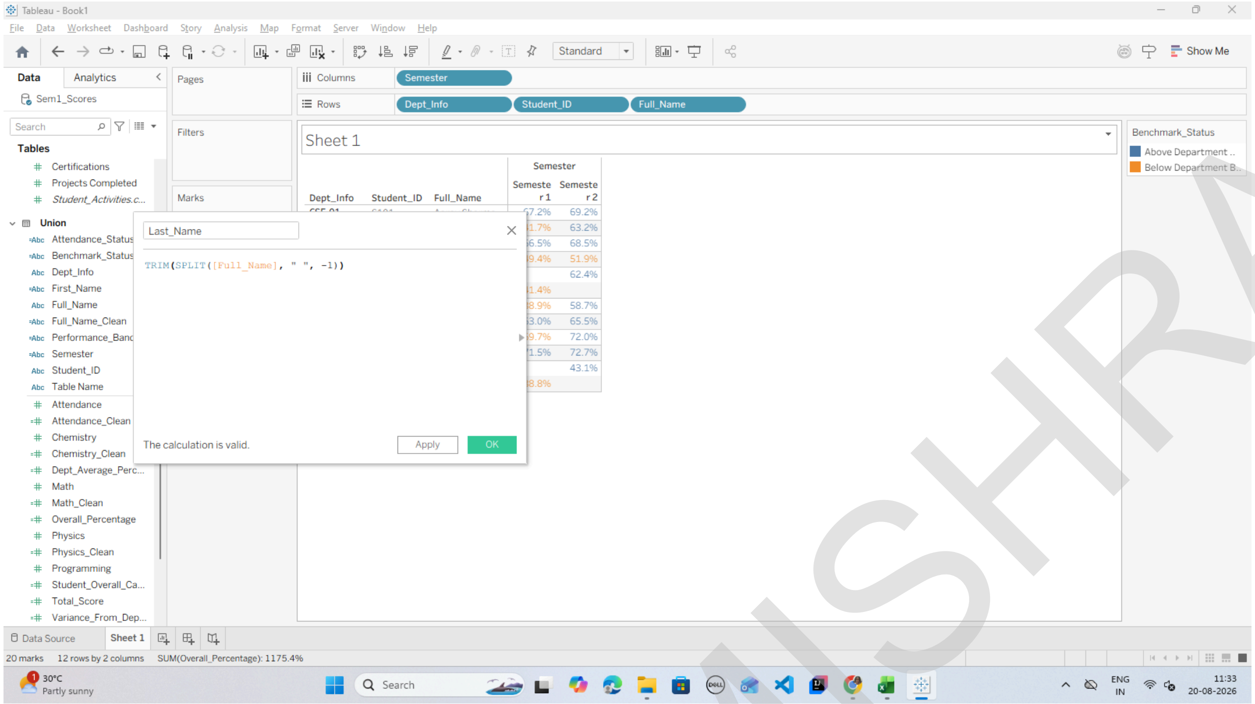
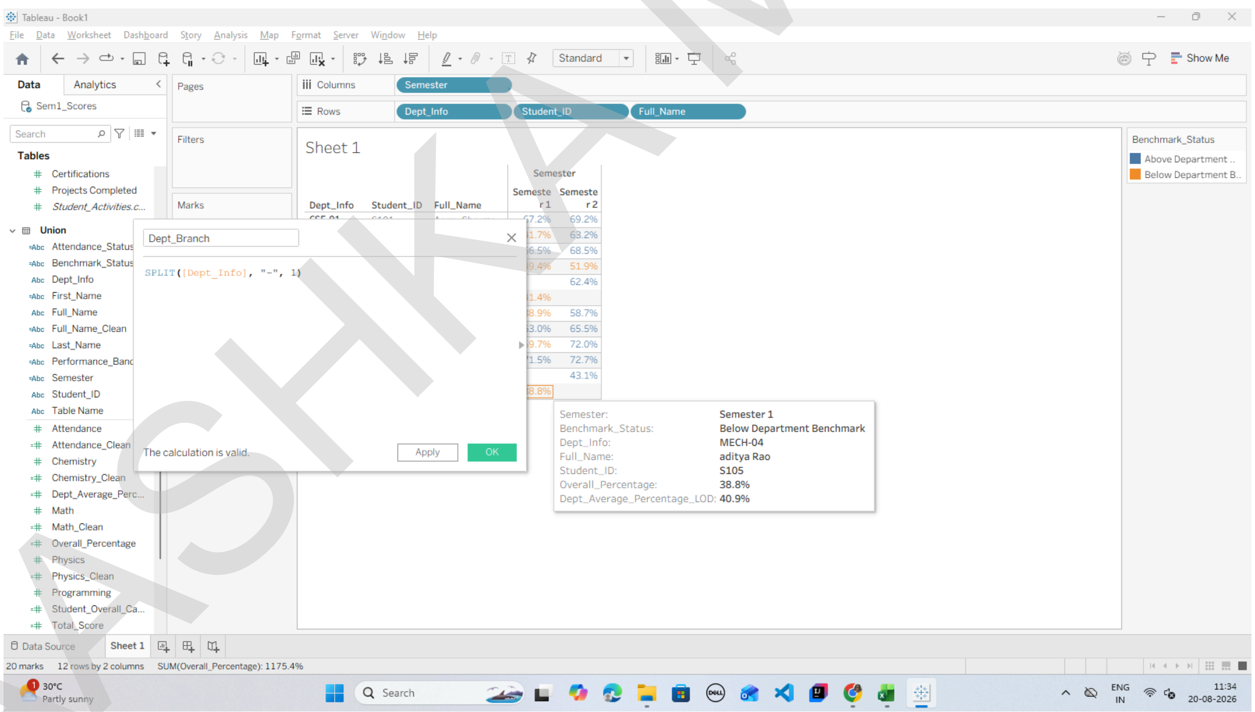
SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: Data Visualization with Power BI and Tableau



Step 3: Split Composite Department Codes (Dept_Info)



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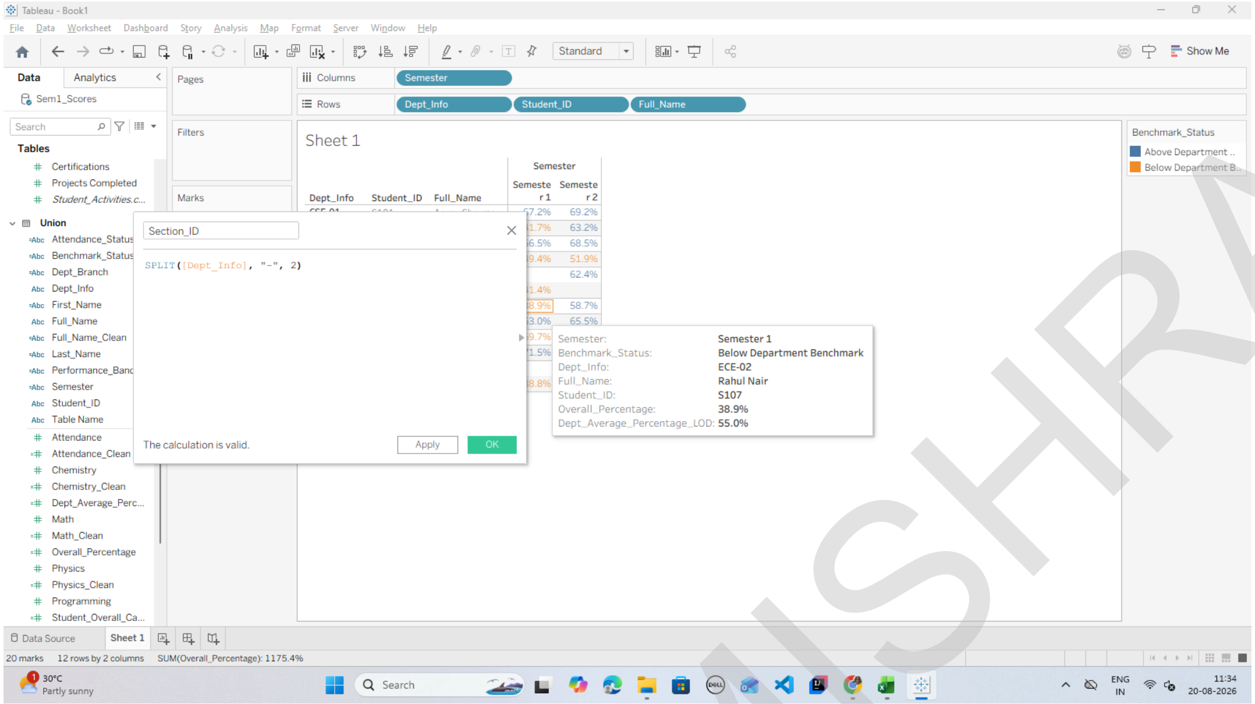
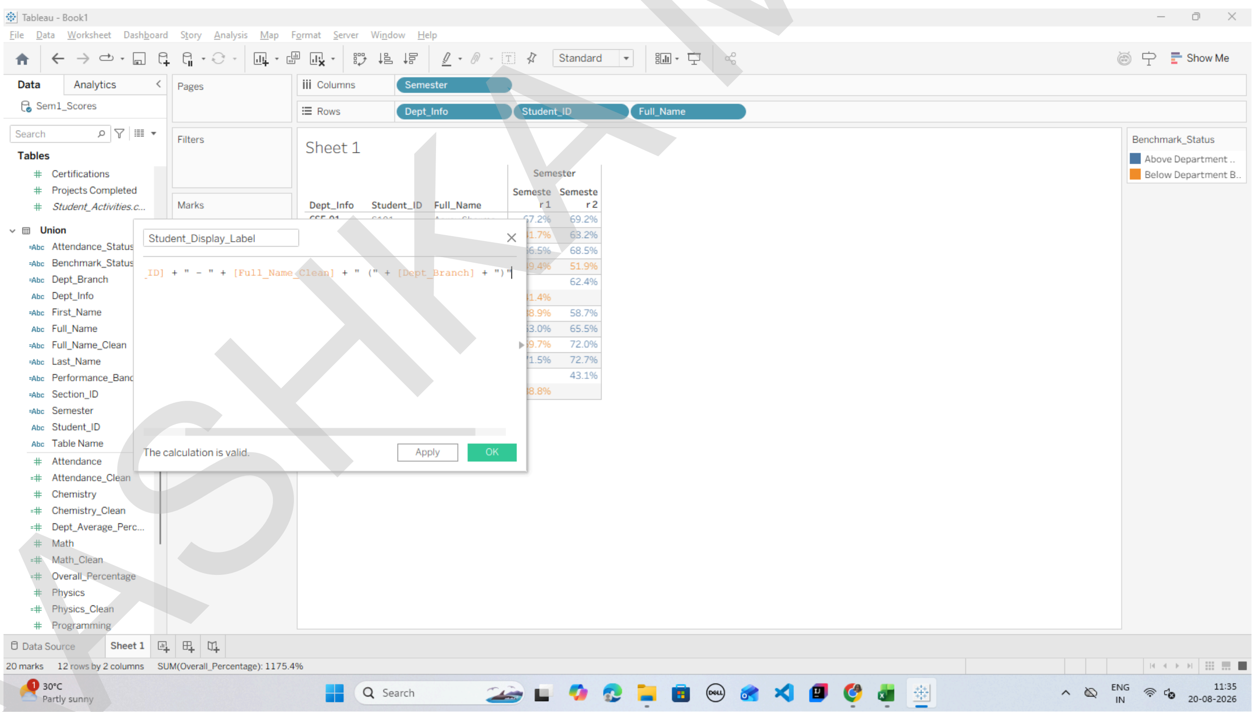


Step 4: Create a Clean Composite Student Label



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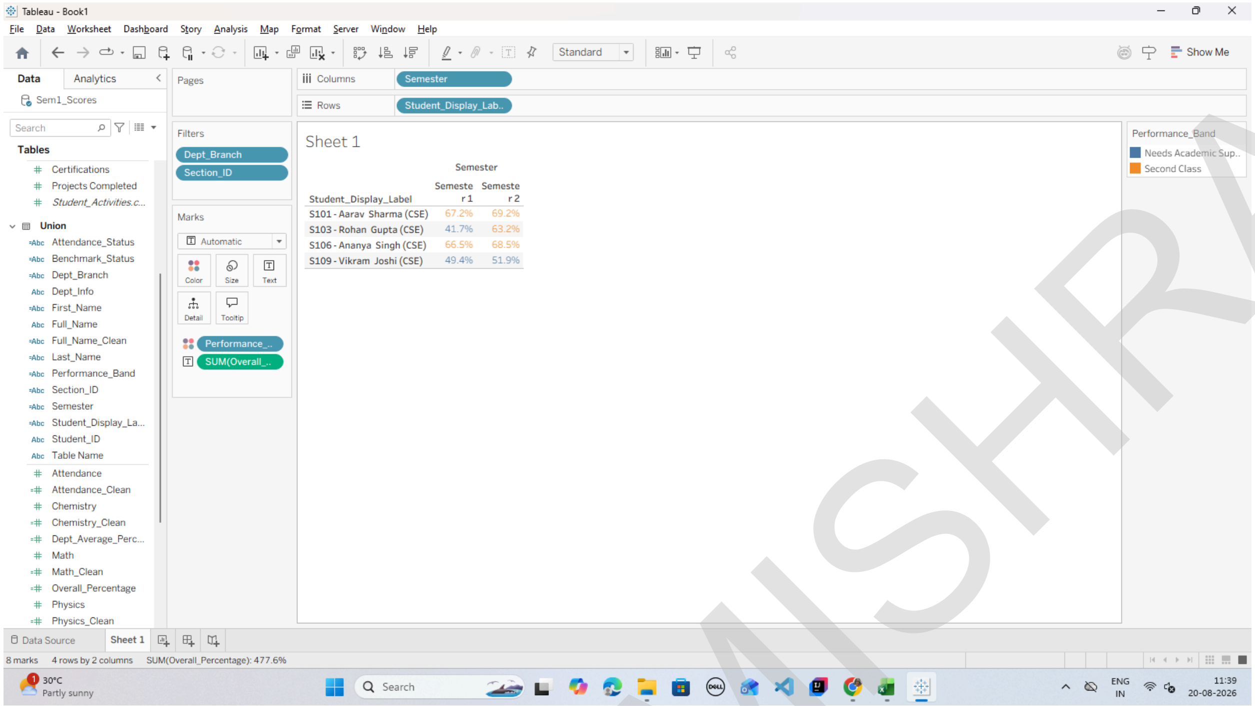
Step 5: Verify in the Final Dashboard View



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